



LEARNING & Instruction

Lecture : 31



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LEARNING: DEFINITIONS

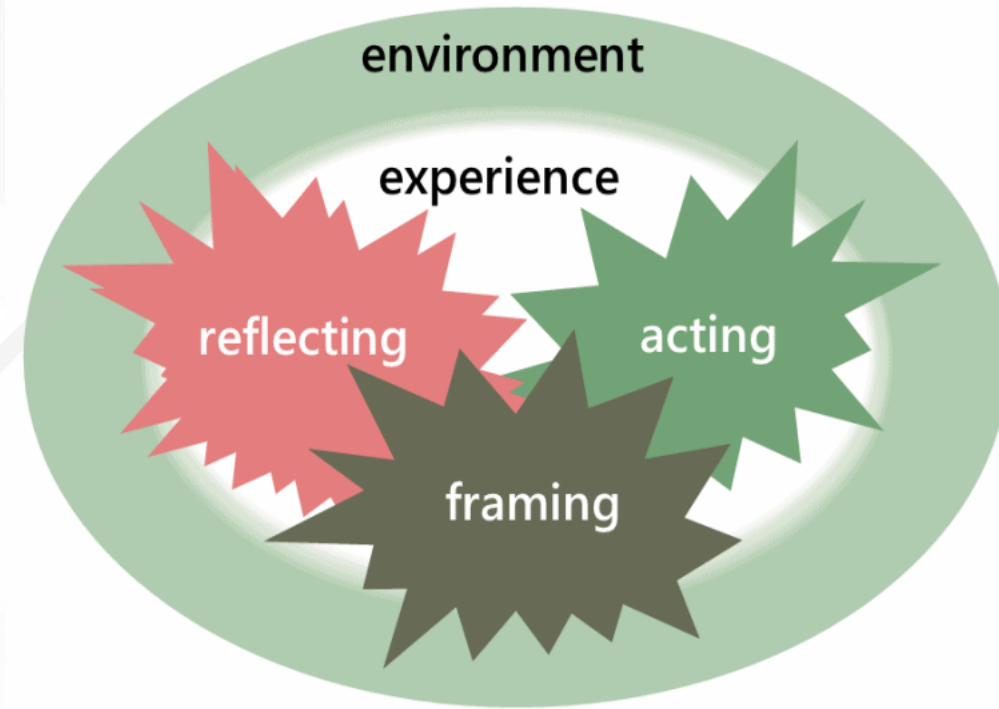
- The cognitive process of acquiring skill or knowledge.
- Simon defined learning as changes in a system that result in improved performance over time on tasks similar to those done previously.
- A process of ADAPTATION by which a set of adjustable parameters is automatically modified so that some objective is more readily achieved.
- A relatively permanent change in cognition, resulting from experience and directly influencing behavior.
- Learning is thought to be the act or process which develops or changes the behaviour of the learner to a degree of permanence, usually with the intervention of an educator.



- Benjamin S. Bloom's **Taxonomy of Educational Objectives** –
- **Cognitive**: mental skills (knowledge).
- **Affective**: growth in feelings or emotional areas (attitude or self).
- **Psychomotor**: manual or physical skills .
- **Learning as a process-**
- David Kolb's exploration of **experiential learning**
- concrete experience,
- observation and reflection,
- the formation of abstract concepts, and
- testing in new situations.

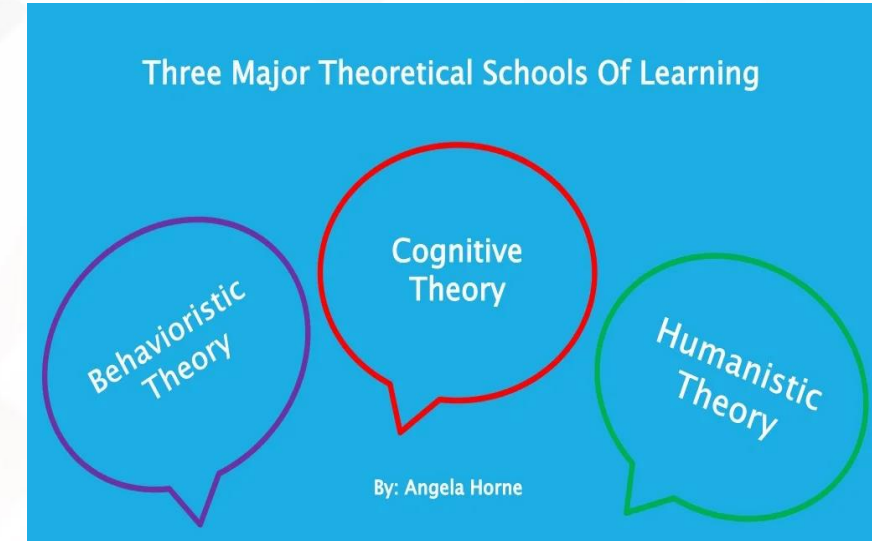


- John Dewey's Reflection, learning and education:
- Returning to experience – that is to say recalling or detailing salient events.
- Attending to (or connecting with) feelings – this has two aspects: using helpful feelings and removing obstructive ones.
- Evaluating experience – this involves re-examining experience in the light of one's intent and existing knowledge etc.
- It also involves integrating this new knowledge into one's conceptual framework.



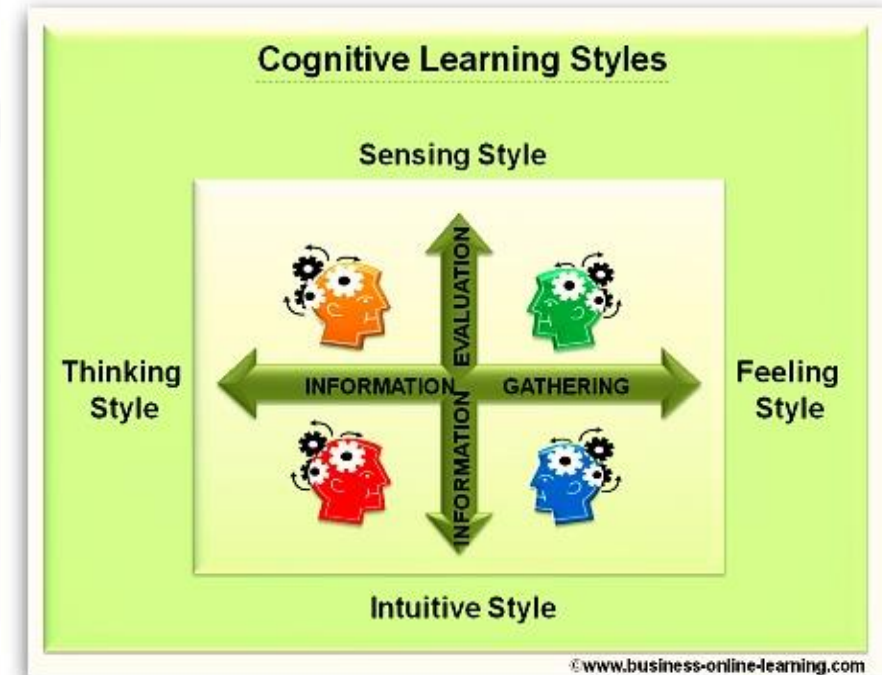
- **Schools of Learning Theories-**

- The behaviorist orientation to learning
- The cognitive orientation to learning
- The humanistic orientation to learning
- The social/situational orientation to learning
- The constructivist/social constructivist orientation to learning.
- **The Behaviorist orientation to learning-key** assumptions
- Observable behavior rather than internal thought processes are the focus of study.
- In particular, learning is manifested by a change in behavior.
- The environment shapes one's behavior; what one learns is determined by the elements in the environment, not by the individual learner.
- The principles of contiguity (how close in time two events must be for a bond to be formed) and reinforcement (any means of increasing the likelihood that an event will be repeated) are central to explaining the learning process.



• The Cognitive orientation to learning- Principles

- Instruction should be well-organized; **Well-organized materials** are easier to learn and to remember.
- **Instruction should be clearly structured**; Subject matters are said to have inherent structures – **logical relationships** between **key ideas** and **concepts** – which link the parts together.
- The **perceptual features of the task are important**; **Learners attend selectively** to different aspects of the environment;
- Thus, **the way a problem is displayed** is important if learners are to understand it.
- **Prior knowledge is important**; Things must fit with what is already known if it is to be learnt.
- Differences between individuals are important as they will affect learning; or methods of approach influence learning, **Differences in 'cognitive style'**
- **Cognitive feedback** gives **information to learners** about their **success or failure** concerning the task at hand;
- **Reinforcement** can come through giving information – a '**knowledge of results**' – rather than simply a reward.





Approaches In Education

- **Open Classroom**
-Student centered with focus on individual growth, critical thinking, commitment to lifelong learning.
-Not age/grade specific nor curriculum bound
- **Learning Styles**
-Allow student to use a learning style that suits them.
- **Co-operative Learning**
-Students engage with one in other to work towards a common goal
- **Experimental Learning**
-Learning from experience that addresses the needs and wants of the learner



- **Humanistic orientations to learning**-Significant learning elements (Carl Rogers,1983)
- It has a **quality of personal involvement**—the whole person in **both feeling** and **cognitive aspects being in the learning** event.
- It is **self-initiated**; Even when the impetus or stimulus comes from the outside, the **sense of discovers of reaching out**, of **grasping** and **comprehending**, comes **from within**.
- It is pervasive; It makes a difference in the behavior, the **attitudes**, perhaps even the **personality of the learner**.
- It is evaluated by the learner;
- She knows whether it is meeting her need, whether it leads toward what she wants to know, **whether it illuminates the dark area of ignorance she is experiencing**.
- The **locus of evaluation**, we might say, resides definitely in the learner.
- Its **essence is meaning**; When such learning takes place, the **element of meaning** to the learner is built into the **whole experience**.





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- **The Social/Situational orientation to learning-**
- Attending to a behavior; remembering it as a possible model or paradigm; and playing out how it may work for them in different situations (rehearsal) are key aspects of observational learning(Bandura,1977).
- Situated learning –Lave and Wenger (1991); Learning is not just the acquisition of certain forms of knowledge, but includes social relationships – situations of co-participation depends on two claims:
 - It makes no sense to talk of knowledge that is decontextualized, abstract or general.
 - New knowledge and learning are properly conceived as being located in communities of practice;
 - Learning is in the relationships between people;
 - Educators work so that people can become participants in communities of practice – they need to explore with people in communities how all may participate to the full.
 - There is an intimate connection between knowledge and activity – Learning is part of daily living such as Problem solving and learning from experiences .





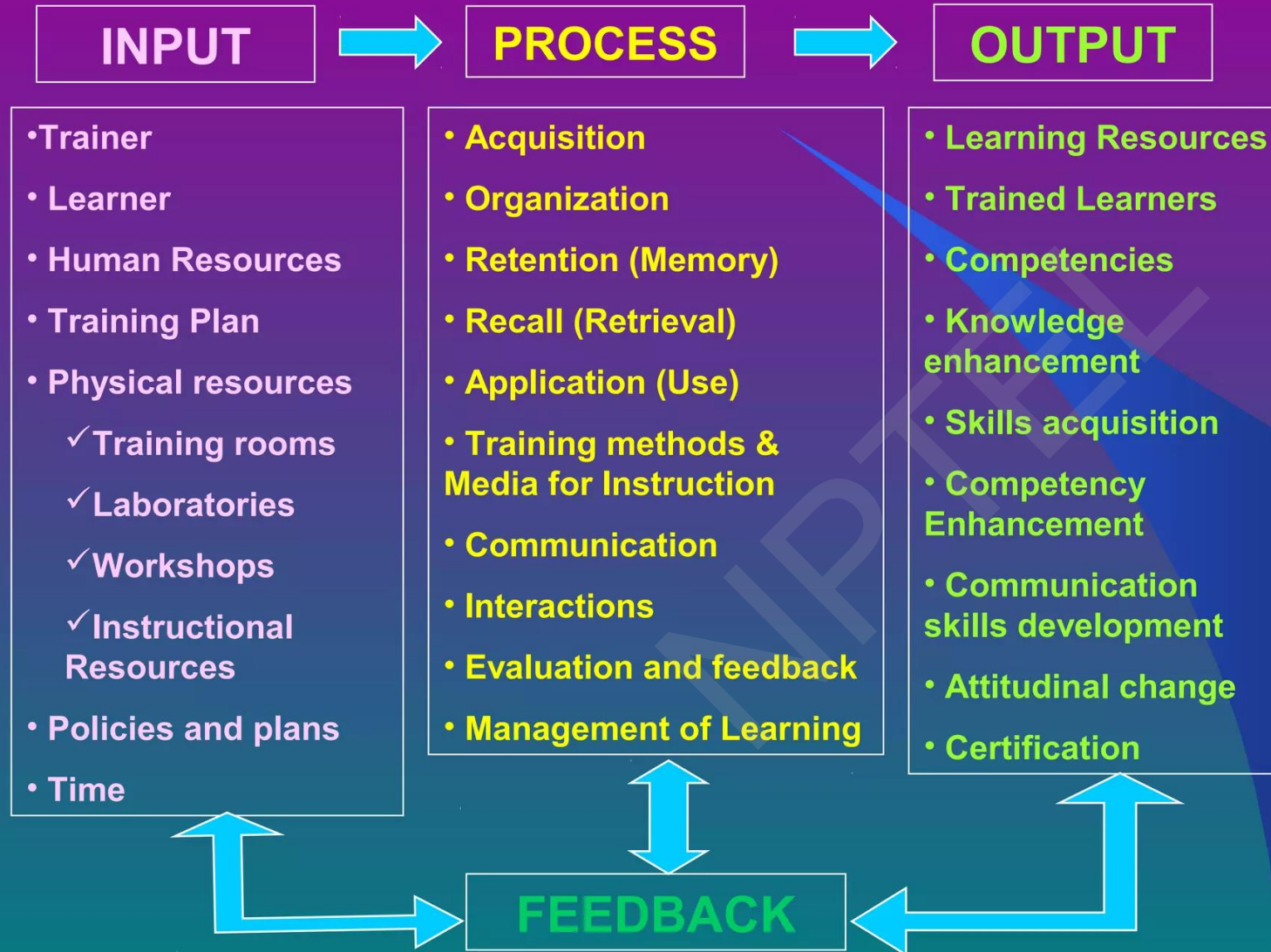
• The Constructivist / Social constructivist orientation to learning-

- Learning as a mental activity— knowledge is created; Active learning
- Social, meaning created by each learner (personal)”
- Learners do not transfer knowledge from the external world into their memories; rather they build personal interpretations of the world based on individual experiences and interactions”
- “Human beings construct knowledge by giving meaning to current experiences in light of prior knowledge, mental structures, experiences, and beliefs.
- It is based on the assumption that the source of a person’s understanding of external phenomena is in the person’s mind.
- Mind is viewed as an active participant in helping people make sense of reality”
- “Children’s abilities should be assessed under a variety of work conditions; Engagement, participation, social, cultural.
- Constructivist teaching involves students in collaborative groups, such as cooperative learning, reciprocal teaching, and long-term projects.

SOCIAL CONSTRUCTIVISM



Process of Learning





Factors Affecting Learning

1. Individual Variables

TRAINING PROCESS

2. Task (Learning) Variables

3. Environmental Variables



Factors Affecting Learning

INDIVIDUAL VARIABLES

- LEARNING CAPABILITIES
- INTREST
- PERCEPTION
- EXPERIENCES
- ENDURANCE
- PERSONAL FACTORS
- EGO

TASK VARIABLES

- OBJECTIVES OF THE TRAINING
- TRAINER'S COMPETENCE
- VARIETY OF EXPERIENCES
- COMMUNICATION GAP
- RATIONALITY
- TASK OVERLOAD
- INTERVENTIONS

ENVIRONMENTAL VARIABLES

- SPACE
- AMBIENCE
- CONDUSIVE ENVIRONMENT
- DECORUM
- ERGONOMICS
- AMENITIES
- LOCAL VISITS
- SERVICE QUALITY





TYPES OF LEARNING

Learning is the process through which individuals acquire, retain, and apply new knowledge, skills, or behaviors by engaging with and interpreting their experiences.

ACTIVE VS PASSIVE

- **Active learning** – Active learning refers to learning that gets students to learn through doing, such as through experimentation and play.
- **Passive learning** – Passive learning refers to situations where the teacher tells students information and students must passively sit and watch.

SHALLOW VS DEEP

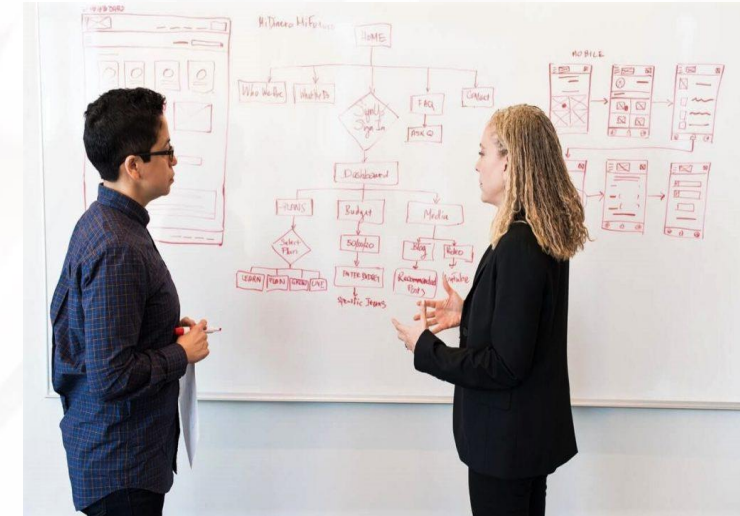
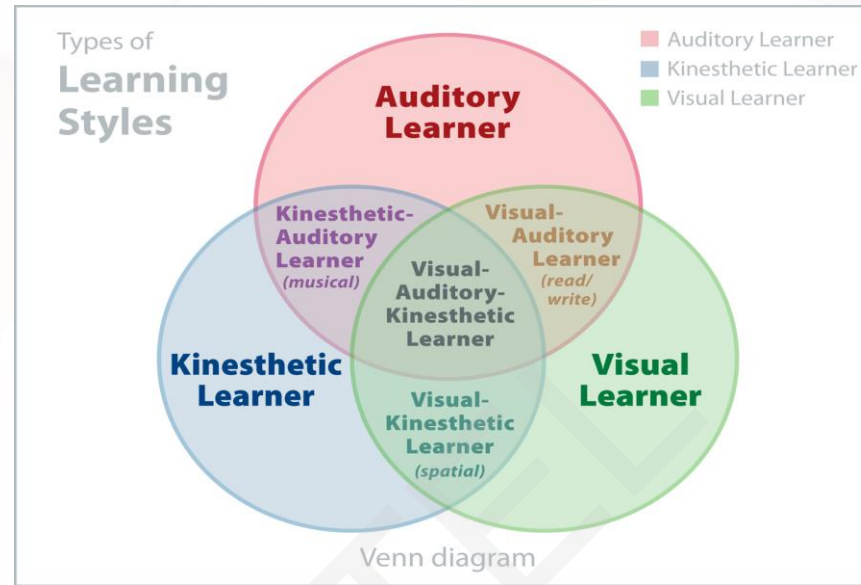
- **Shallow learning** – This involves only learning things at a surface level, such as through memorization rather than true understanding.
- **Deep learning** – This refers to learning that achieves detailed, contextualized, and thorough understanding.

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- Learning Styles- Types-

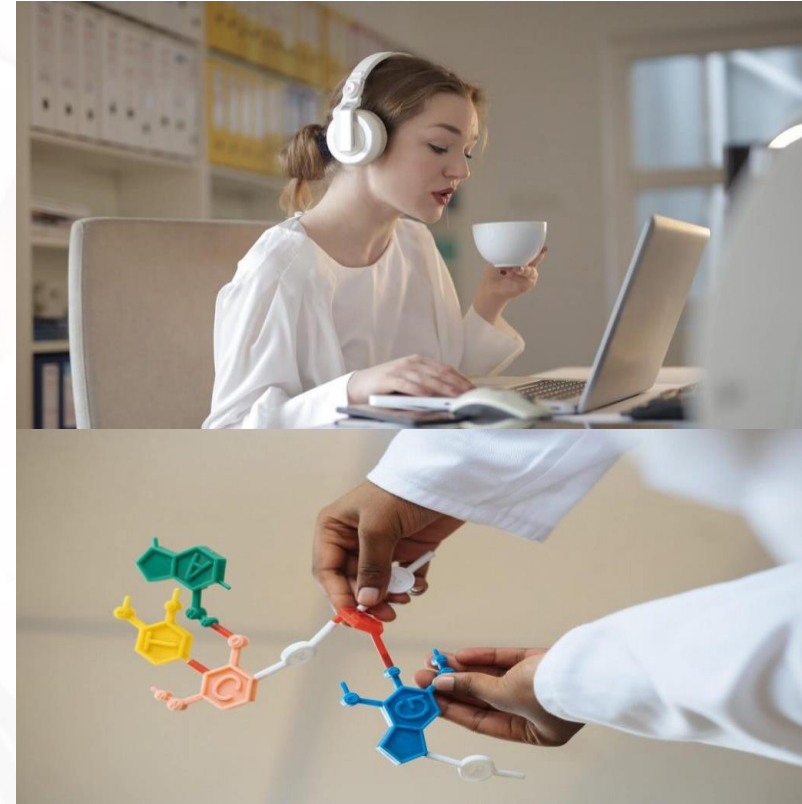
- Visual learning style
- Auditory learning style
- Kinesthetic learning style
- Reading/Writing
- Logical/analytical learners
- Social/linguistic learners
- Solitary learners
- Nature learners



- **Visual learning style**-Visual learners are individuals who prefer to take in their information visually—be that with maps, graphs, diagrams, charts, and different visual aids such as patterns and shapes.
- The best way to present to visual learners is by showing them the relationship between different ideas visually through Flow Chart.



- **Auditory learning style** – This type of individuals learn better when they take in **information in auditory form** when it is **heard or spoken**.
- They are prone to **sorting their ideas after speaking**, rather than thinking ideas through before.
- Thus, **saying things out loud** helps them **understand the concept**.
- Auditory learners learn best when information is presented to them via **strategies** that involve **talking**, such as **lectures** and **group discussions**.
- They can **benefit from repeating back the lessons**, having recordings of the lectures, group activities that require classmates explaining ideas, etc.
- **Kinesthetic learning style**- Kinesthetic learners are individuals who **prefer to learn by doing**.
- They enjoy a **hands-on experience**.
- They are usually more in touch with reality and more connected to it, which is why they require using **tactile experience to understand something better**.



- **Reading/Writing**-This type of learners consume **information best when it's in words**, whether that's by writing it down or reading it.
- To them, **text is more powerful** than any kind of visual or auditory representation of an idea.
- These individuals usually **perform very well on written assignments**;
- It would be best to have them **describe charts and diagrams** by written statements, take **written quizzes** on the topics, or give them **written assignments**.
- **Logical/analytical learners**- This type of learners depend on logic and **analytical skills to understand a particular subject**.
- They search for **connections, causes, patterns**, and results in their learning.
- A teacher can engage and **motivate analytical learners by** posing **questions that require interpretation**, using material that **activates problem-solving skills** and stimulating students to reach **conclusions based on facts or reasoning**.





- **Social/linguistic learners**-These types of learners favor educational lessons that include peer work or participation.
- Social/ linguistic learners get two things out of this participation: socializing (which they love) and a better understanding of a subject.
- Teachers can motivate these types of learners by using role-playing, group activities, and encouraging student interaction (asking questions, sharing stories, etc.).
- **Solitary learners**-Otherwise known as solo learners, these students are the opposite of social learners.
- Solitary learners prefer to study alone without having to interact with other learners.
- Individual work is a solo student's forte.
- Teachers can help these types of learners by using activities that require individual work (including keeping a diary) and problem-solving skills, recognizing a student's individual accomplishments, etc.



- **Nature learners**-These types of learners excel when in **contact with nature**.
- A nature learner's **ideal study environment** is a calm and **relaxing** environment.
- If we had to compare nature learners with another type, it would be **tactile learners**.
- The only difference is the nature part of this deal, as nature learners **need to be outside to learn better**.
- While learning in nature may not always be possible, teachers can still nurture this learning style in students by **assigning hands-on activities, having classes outdoors** when possible, and **using nature examples** when explaining a new lesson.

FACTORS OF LEARNING	
Principle Learning Factors	Percent Improvement
1. Aligning learning and performance contexts	10 to 55%
2. Providing retrieval practice and testing	30 to 100%
3. Using appropriate feedback schedules	15 to 50%
4. Prompting meaningful repetitions	30 to 110%
5. Spacing repetitions and practice over time	5 to 40%
6. Varying materials and delivery methods	not yet determined
7. Avoiding the addition of distracting elements	25 to 50%
8. Appropriately using learning objectives	5 to 45%





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- **Instruction refers** to those curriculum-related, professionally-informed decisions that teachers purposefully enact to enhance learning opportunities for students.
- Effective instruction(EI) is **interactive** and designed to **accommodate student learning needs** and **styles** through a variety of teaching practices.
- EI is guided by general **pedagogical approaches** and specific **instructor practices**.
- Effective instruction is eclectic.
- EI is tied directly to the success of the **learning experience**.
- EI is empowered **professional practice in action**.
- EI **integrates the components** of the Core Curriculum.
- EI is **generative** and **dynamic**.
- EI recognizes there is an **art** as well as a **science to teaching**.
- EI acknowledges a **comprehensive understanding of the instructional cycle**.
- EI finds best expression when **educators collaborate** to develop, implement, and **refine their professional practices**.



INSTRUCTION

INSTRUCTION IS A SET OF EVENTS OUTSIDE THE LEARNER TO FACILITATE THE INTERNAL PROCESS OF LEARNING.



Events of Instruction

S.No	INSTRUCTIONAL EVENTS	RELATION TO INTERNAL LEARNING PROCESS
1.	GAINING ATTENTION	RECEPTION OF PATTERNS OF NEURAL IMPULSES
2.	INFORMING THE LEARNER OF THE OBJECTIVES	ACTIVATING THE PROCESS OF EXECUTIVE CONTROL
3.	STIMULATING RECALL OF PREREQUISITE LEARNING	RETRIEVAL TO WORKING MEMORY
4.	PRESENTING THE STIMULUS MATERIAL	EMPHASIZING FEATURES OF SELECTIVE PERCEPTION
5.	GUIDING LEARNING	SEMANTIC ENCODING
6.	ELICITING THE PERFORMANCE	ACTIVATING RESPONSE ORGANIZATION
7.	PROVIDING FEEDBACK ABOUT PERFORMANCE CORRECTNESS	ESTABLISHING REINFORCEMENT
8.	ASSESSING THE PERFORMANCE OF STUDENTS	ACTIVATING RETRIEVAL MAKING REINFORCEMENT POSSIBLE
9.	ENHANCING RETENTION AND TRANSFER	PROVIDING CUES & STRATEGIES FOR RETRIEVAL

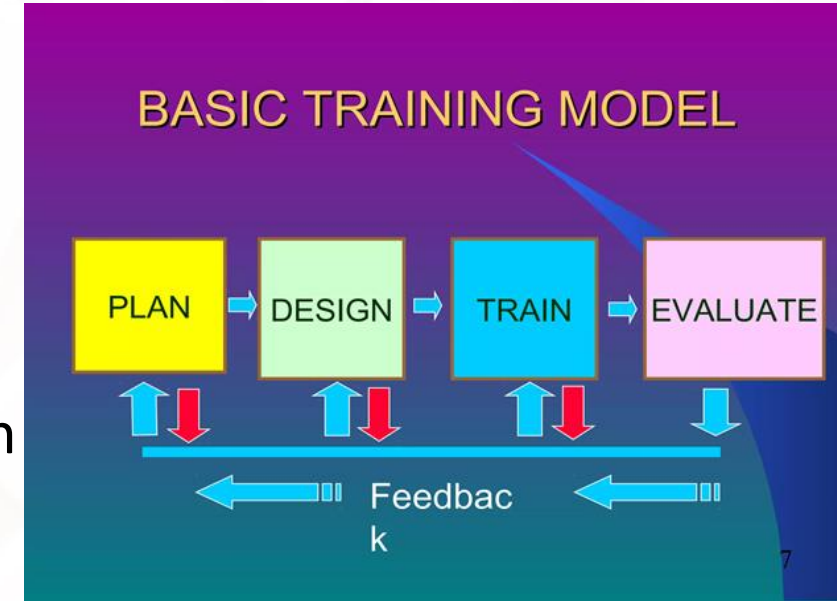


The Role of the Trainer

- The Expert Role: transmits information about a subject to an audience.
- The Planning Role: designs or plans the learning experience or the learning environment to build sustainable teams.
- The Instructor: guides or directs the learning situation often telling the learner what to do
- The Facilitator: responds to the learner's needs and lends guidance and support to facilitate self-learning.
- Resource Person: provides materials and information to the learners on the identified gaps.
- The Co-Learner: learns along side the learners. Mutually plans the learning objectives and activities.
- The Model Role: models or influences behaviour and values.



- **Different types of instruction:**
- **Direct instruction:** **teacher-centered**, with little to no input from students. The teacher gives instruction, demonstrations, and explanations.
- **Indirect instruction:** **student-centered**, with more input from students. The teacher guides students to discover concepts, principles, and skills.
- **Interactive instruction:** **group-centered**, with more interaction among students and teacher. The teacher facilitates discussions, debates, role-plays, and cooperative learning.
- **Independent instruction:** **individual-centered**, with more autonomy for students. The teacher assigns projects, research, self-study, and homework.
- **Experimental instruction:** **inquiry-centered**, with more hands-on learning for students. The teacher engages students in experiments, simulations, problem-solving, and investigations.



- **Principles of Instruction-** (Merrill (2002))
- **1. Task Related or Problem Centered:** Learning is promoted when learners are engaged in solving real-world problems.
- **2. Activation:** Learning is promoted when existing knowledge is activated as a foundation for new knowledge.
- **3. Demonstration:** Learning is promoted when new knowledge is demonstrated to the learner.
- **4. Application:** Learning is promoted when new knowledge is applied by the learner.
- **5. Integration:** Learning is promoted when new knowledge is integrated into the learner's world.



Fig. 1.2: Merrill's First Principles of Instruction



- **Instructional System Design: ADDIE Model**
- Instructional Systems Design/ISD is used for **assessing** and **developing courses** and solutions needed for **formal training**.
- The operative word here is “**systematic**,” as ISD is based on a framework and systems **approach to designing courses for training delivery**.
- Applying ISD in designing and developing a course keeps Instructional Designers (IDs) on track for a **learners-based** rather than teacher-based **approach to training**.
- This systematic approach ensures an **effective learning** process **grounded on learning objectives**.
- The most popular ISD is the **ADDIE** model, which stands for **Analysis, Design, Development, Implementation, and Evaluation**.
- ADDIE is a **step-by-step framework** used by instructional designers, training developers, and learning and development specialists **to create educational or training programs**.
- ADDIE is an instructional design model that aims **to organize content, streamline development**, and ensure a **structured framework** for every course.



- **Principles of Instructional Systems Design-**
- **It should move at an efficient pace.**
- An instructional program should **be able to teach a subject matter efficiently**, given that participants normally require several hours of learning.
- It should **encourage its learners** to cover the learning activities whenever they can, enabling them **to apply the subject matter in practical situations** quickly.
- To do this- **Instructor should-**
- 1) **Look into the students' background** and **learning capabilities**.
- 2) Add in **supplementary resources** and extra (remedial) materials to allow students to **absorb the lessons faster**.
- 3) Use a clear, straightforward, and **organized format** while giving structure to the build-up of knowledge, such as referring back to previous lessons and **explaining how the current subject links to the next one**.
- 4) **Break down the lessons into smaller chunks** instead of giving a significant amount of information in one go.



An organized procedure that includes the steps of Analyzing, Designing, Developing, Implementing, and Evaluating instruction (Seels & Richy, 1994).





- **It should put information into context.**
- Enable the learners to understand the subject more deeply by **linking new information** and theories to their **existing knowledge**.
- Provide context as to **how they can apply it in real life**. Use several examples and mix up the formats, i.e. written and practical.
- **It should be community-based.**
- Even when the training program is remote, **learning should take place within a community of learners**.
- Provide avenues for **peer feedback** and **student-to-student interactions**.
- Enable teamwork by providing learning activities that require them to **collaborate with others**.
- **It provides opportunities for learners to create original output.**
- Learning does not just involve cognitive functions such as remembering and memorizing.
- **Learning improves** when students are **able to produce output** that **applies** what they have learned



- For example, writing reaction papers or delivering oral presentations help students digest information in a more in-depth manner.
- Critical-thinking and problem-solving exercises help learners understand how theories can be applied in practical situations.
- Working outside of the class, like on-the-job training, will help them see various viewpoints and appreciate how the course becomes significant in day-to-day work.
- It creates and uses evaluation tools implemented at the learner's pace.
- Ensure that learners have a full grasp of the lessons before conducting an evaluation.
- Provide ample time for learning, versus cramming a heavy amount of information and then administering tests at the end of the module.
- Space out the delivery of materials and provide informal evaluations and feedback in small chunks.
- Make sure tests are relevant and effective by evaluating testing instruments to see if they actually work.

Definition of ISD

Instructional Systems Design (ISD)

is the systematic approach to the Analysis, Design, Development, Implementation, and Evaluation of learning materials and activities

In short,

ISD is the process to determine the

5W1H of Learning

(Who, What, When, Where, Why, and How)

Designed & Developed by YY Wang * version 2012 *

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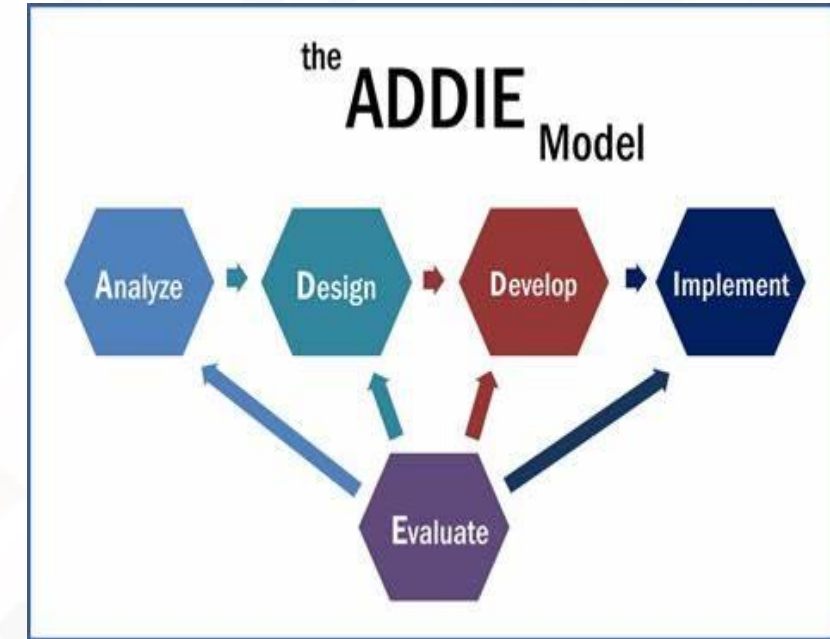
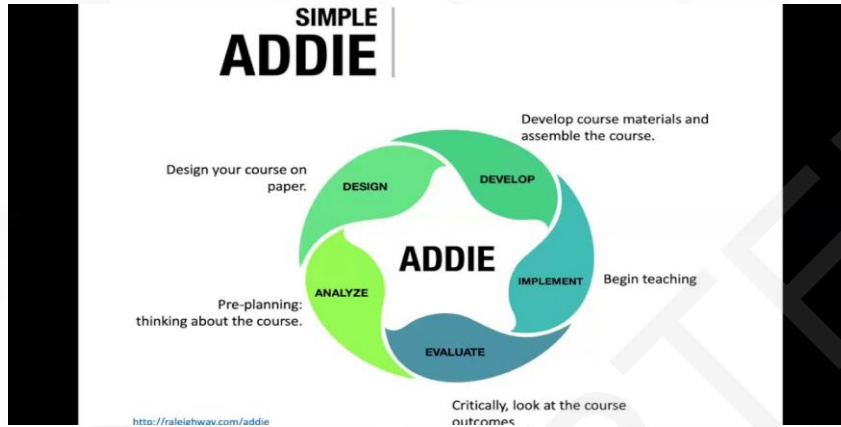
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- ADDIE is a popular model for instructional design, providing a framework to produce effective learning and development programs.
- “ADDIE” itself is an acronym for the five stages of the model:
 - Analyze
 - Design
 - Development
 - Implementation
 - Evaluation
- These five clearly defined processes help ensure training programs and their resources have the required elements to succeed.
- ADDIE methodologies offer a systems-based, iterative learning and development design strategy while incorporating feedback and review to improve.
- There are also other models to apply a systematic process to content development, planning, and developing their corporate training.





E Evaluation

Determining the performance of the training program and what could be improved

A Analysis

Setting goals, researching the target audience, and understanding required resources



I Implementation

The actual delivery of the learning and development program

D Design

Finding a learning solution that aligns with the identified objectives

D Development

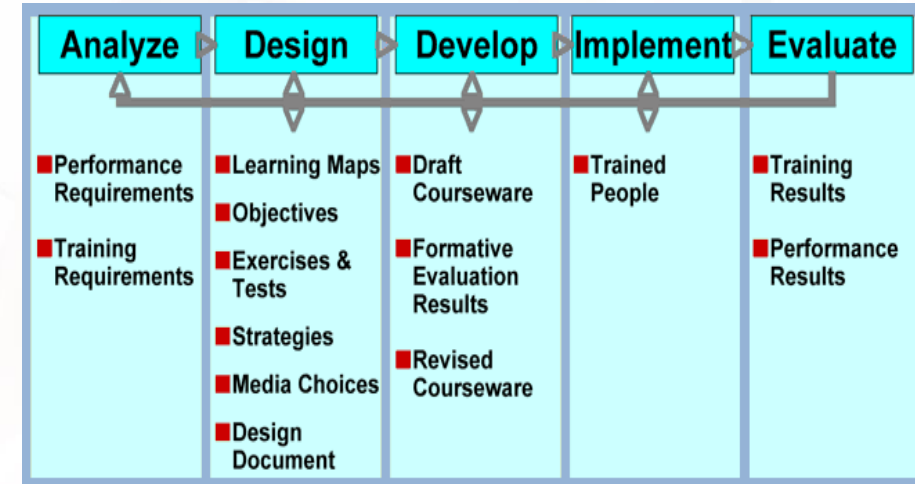
Creating the assets required and testing various methodologies for delivering the content



- In the ADDIE model, each step has an outcome that feeds into the subsequent step.
- **Analysis > Design > Development > Implementation > Evaluation**

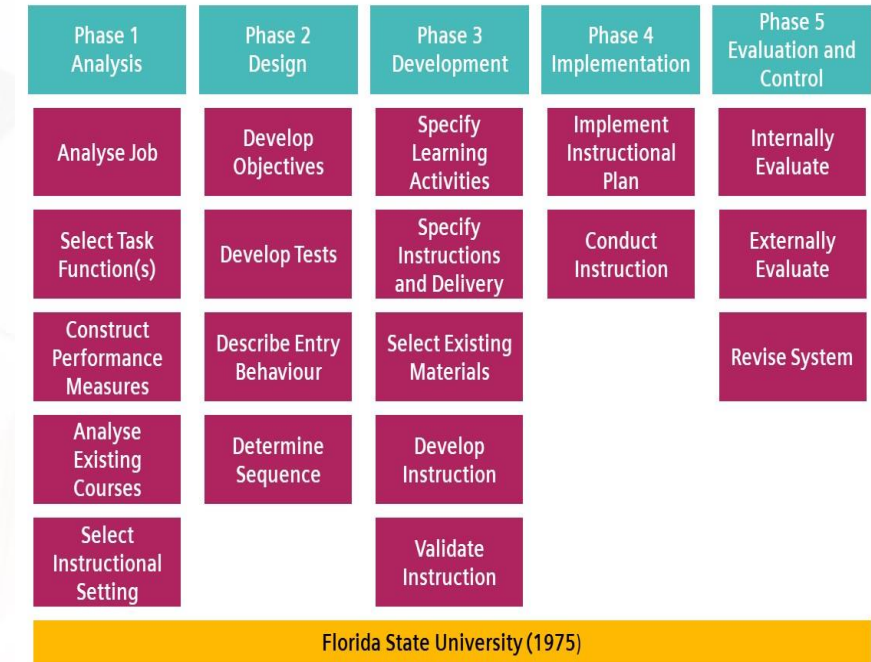
- **Analysis Phase**

- In the analysis phase, instructional problem is clarified, the instructional goals and objectives are established and the learning environment and learner's existing knowledge and skills are identified.
- The following **questions are addressed** during the analysis phase:
 - * Who is the audience and their characteristics?
 - * Identify the new behavioral outcome?
 - * What types of learning constraints exist?
 - * What are the delivery options?
 - * What are the online pedagogical considerations?
 - * What is the timeline for project completion?



• Design Phase

- The design phase deals with learning objectives, assessment instruments, exercises, content, subject matter analysis, lesson planning and media selection.
- The design phase should be systematic and specific.
- Systematic means a logical, orderly method of identifying, developing and evaluating a set of planned strategies targeted for attaining the project's goals.
- Specific means each element of the instructional design plan needs to be executed with attention to details.
- These are steps used for the design phase:
 - * Documentation of the project's instructional, visual and technical design strategy
 - * Apply instructional strategies according to the intended behavioral outcomes by domain (cognitive, affective, psychomotor).
 - * Create storyboards
 - * Design the user interface and user experience
 - * Prototype creation
 - * Apply visual design (graphic design)

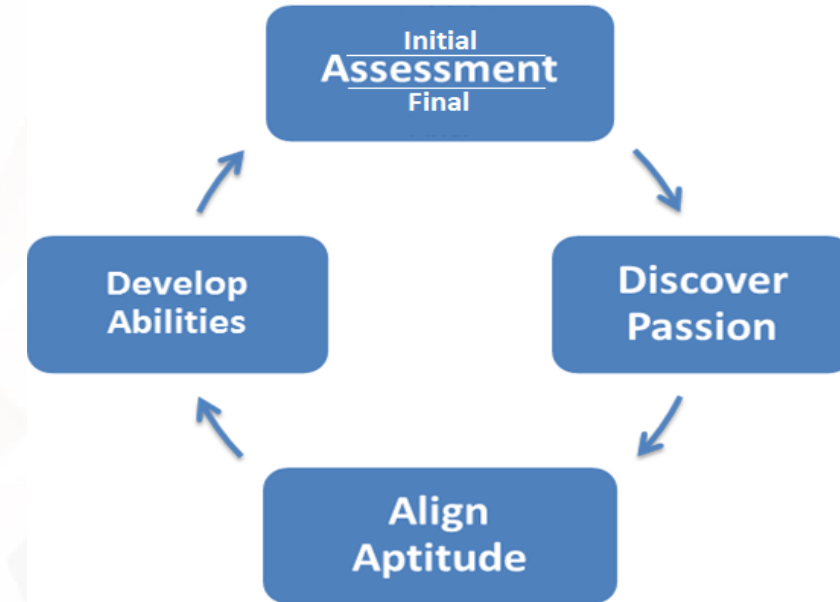


• Development Phase

- The development phase is where the developers **create** and **assemble the content assets** that were created in the design phase.
- Programmers work **to develop** and/or **integrate technologies**. Testers perform debugging procedures.
- The project is reviewed and **revised according to any feedback given**.

• Implementation Phase

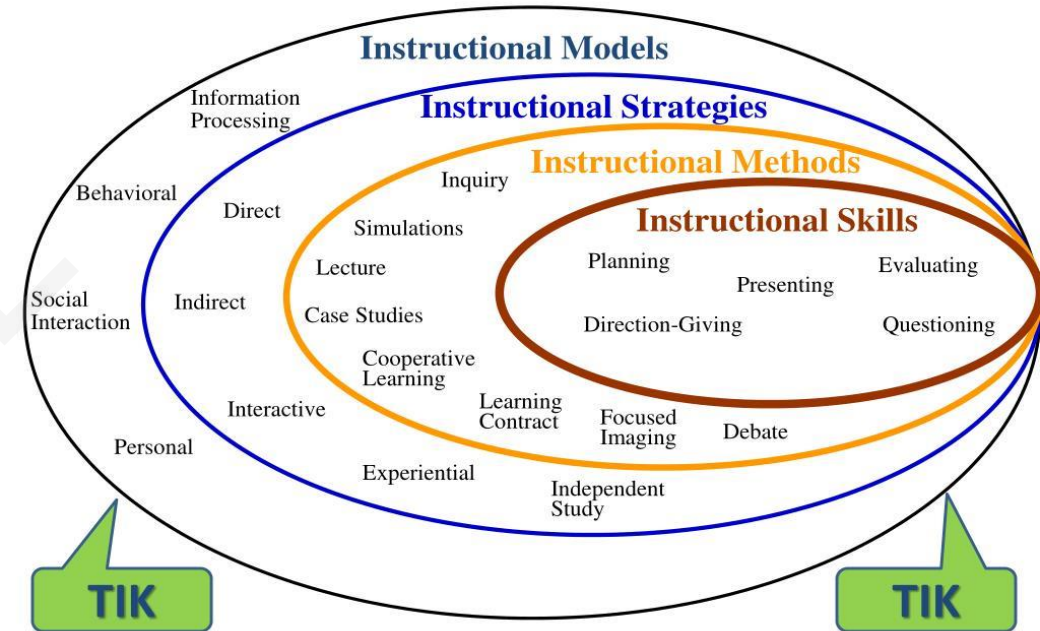
- During the implementation phase, a **procedure for training the facilitators** and the learners is developed.
- The **facilitators' training should cover the course curriculum, learning outcomes**, method of delivery, and testing procedures.
- Preparation of the learners include training them on new tools (**software or hardware**), student registration.
- This is also the phase where the project manager ensures that the books, **hands on equipment**, tools, CD-ROMs and **software** are in place, and that the learning application or **Web site is functional**.

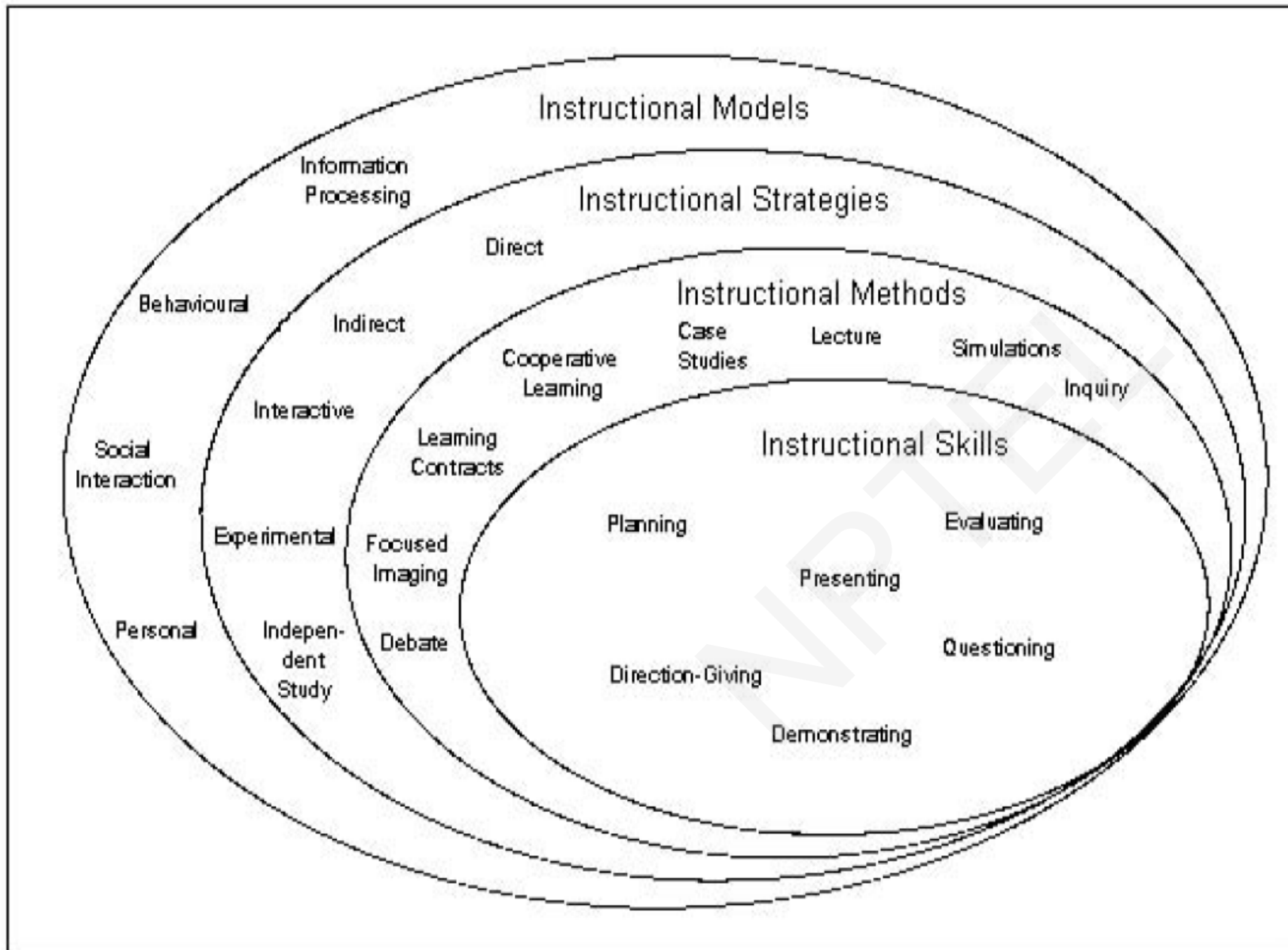


• Evaluation Phase

- The evaluation phase consists of two parts: **formative** and **summative**.
- **Formative evaluation** is present in each stage of the ADDIE process.
- **Summative evaluation** consists of tests designed for domain specific criterion-related referenced items and providing opportunities for feedback from the users.
- **What is Instructional Framework ?**
- The Instructional Framework, identifies and illustrates the **interrelationship among instructional approaches** that, properly used, are acknowledged to be consistent with sound educational practice.
- The approaches are referenced to the **goals of education** and apply to the **objectives of the various curricula**.
- The Instructional model refers to a **broad approach, to methods, strategies and instructional skill**, that represents a specific teaching behavior to achieve expected learning outcomes.

The Instructional Framework







- **Instructional Models-**

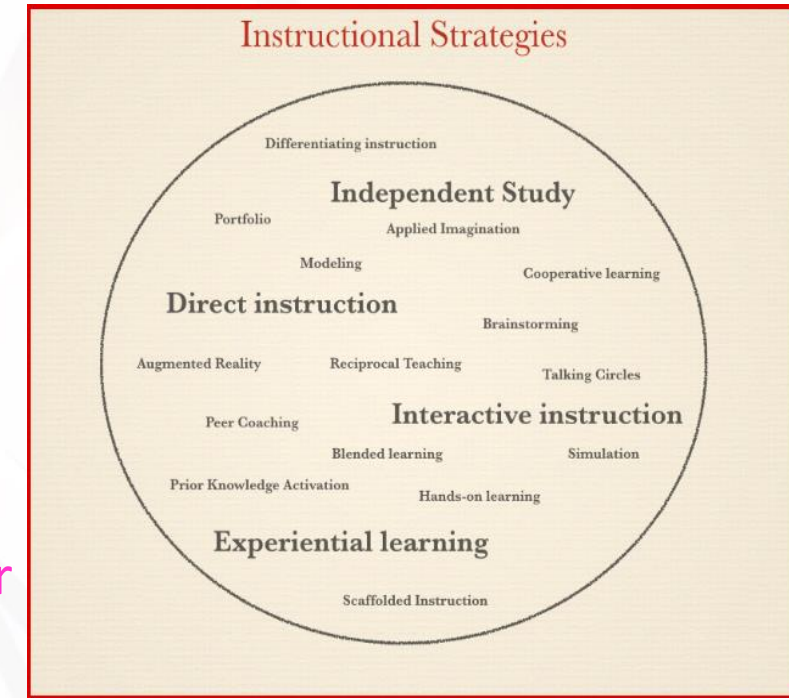
- This represent the broadest **level of instructional practices** and present a philosophical orientation to instruction.
- Models are used to select and **to structure teaching strategies, methods, skills,** and student **activities** for a particular instructional emphasis.
- **Joyce and Weil** (1986) identify **four models: information processing , behavioral, social interaction, and personal.**

- **Instructional Strategies**

- Within each model several strategies can be used.
- **Strategies determine the approach a teacher may take to achieve learning objectives.**
- Strategies can be **classed as direct, indirect, interactive, experiential, or independent.**

- **Instructional Methods**

- Methods are used by teachers **to create learning environments** and to specify the nature of the activity in which the teacher and learner will be involved during the lesson.
- While particular methods are often associated with certain strategies, some methods may be found within a **variety of strategies.**





6 Instructional Strategies

Cooperative Learning Groups

- Can be Ineffective
- Can be Very Effective
- Random Selection of Group Members

Identifying Similarities and Differences.

- Venn Diagrams
- Compare and Contrast
- Metaphors and Analogies

Nonlinguistic Representation/Graphic Organizer

- Visual Imagery
- Auditory Experiences
- Kinesthetic or Whole Body Mode

Homework and Practice

- 10 X Grade = Minutes of Homework a Night.
- Consequence for not Completing Homework.
- Provide Teacher Feedback.

Advanced Organizers

- KWL Charts
- Show How Ideas Relate
- Visual to Help Students Learn

Cues and Questions

- High Level Questions
- Lower Level Questions
- Wait for Students to Answer.
- Cues-Hints

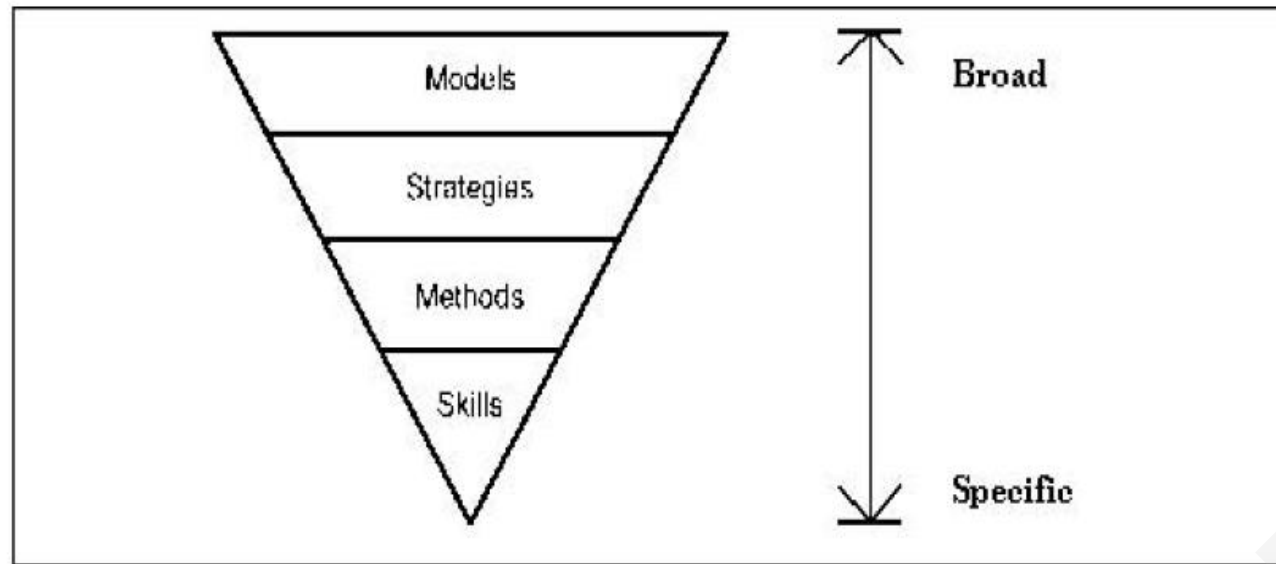
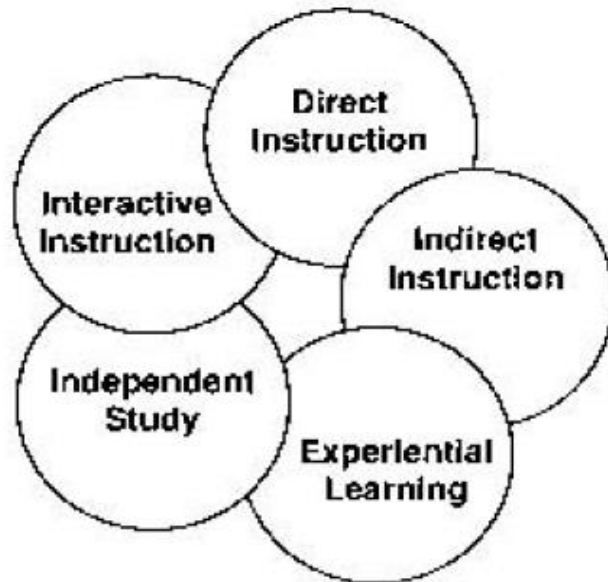


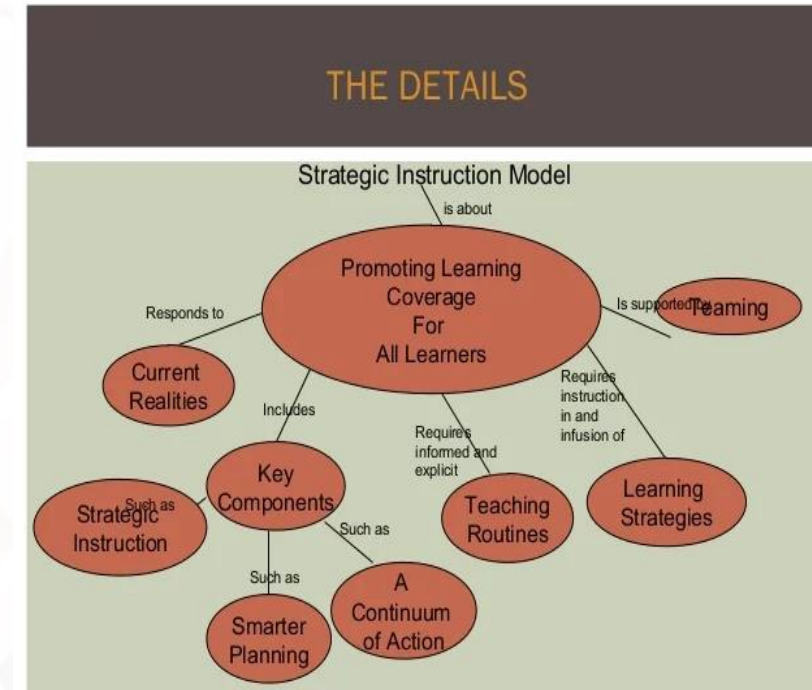
Figure 3. Relationship Among Instructional Models, Strategies, Methods, and Skills



- **Instructional Models** : Joyce and Weil (1986) present four broad models for instruction,
- **Information Processing**
 - This model emphasizes the **acquisition, mastery, and processing information**.
 - The cognitive functioning of the **student is the focus**.
- **Personal**
 - The emphasis in this model is on the **development of the individual self concept**.
 - This involves **development of the processes an individual uses to and organize his or her unique self**.
 - The focus on a **strong, realistic self concept helps to build productive relationships** with others and the environment.
- **Social Interaction**
 - This model emphasizes the **personal and societal relationships** among people.
 - The focus is on improving the student's **ability to relate to others, to engage in democratic processes, and to work productively** in society.

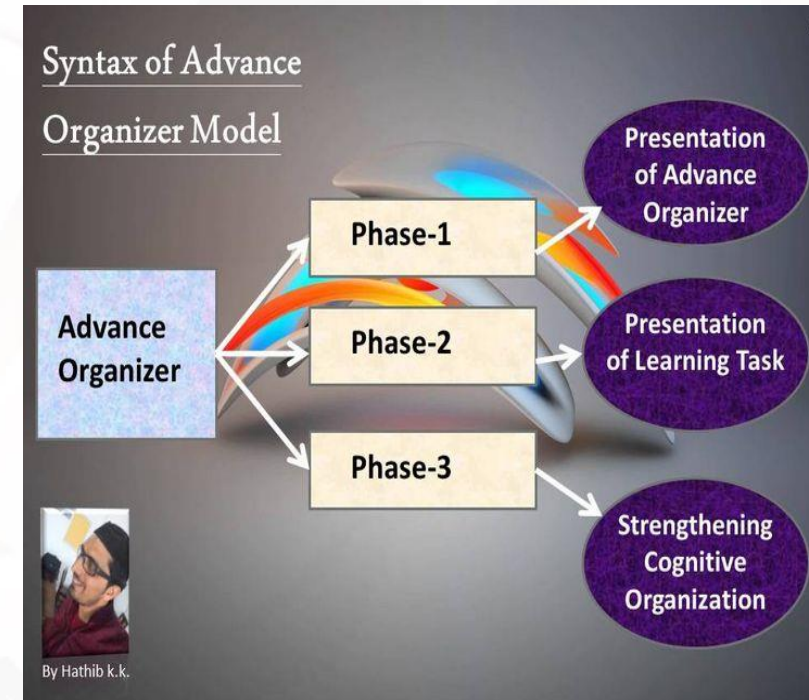


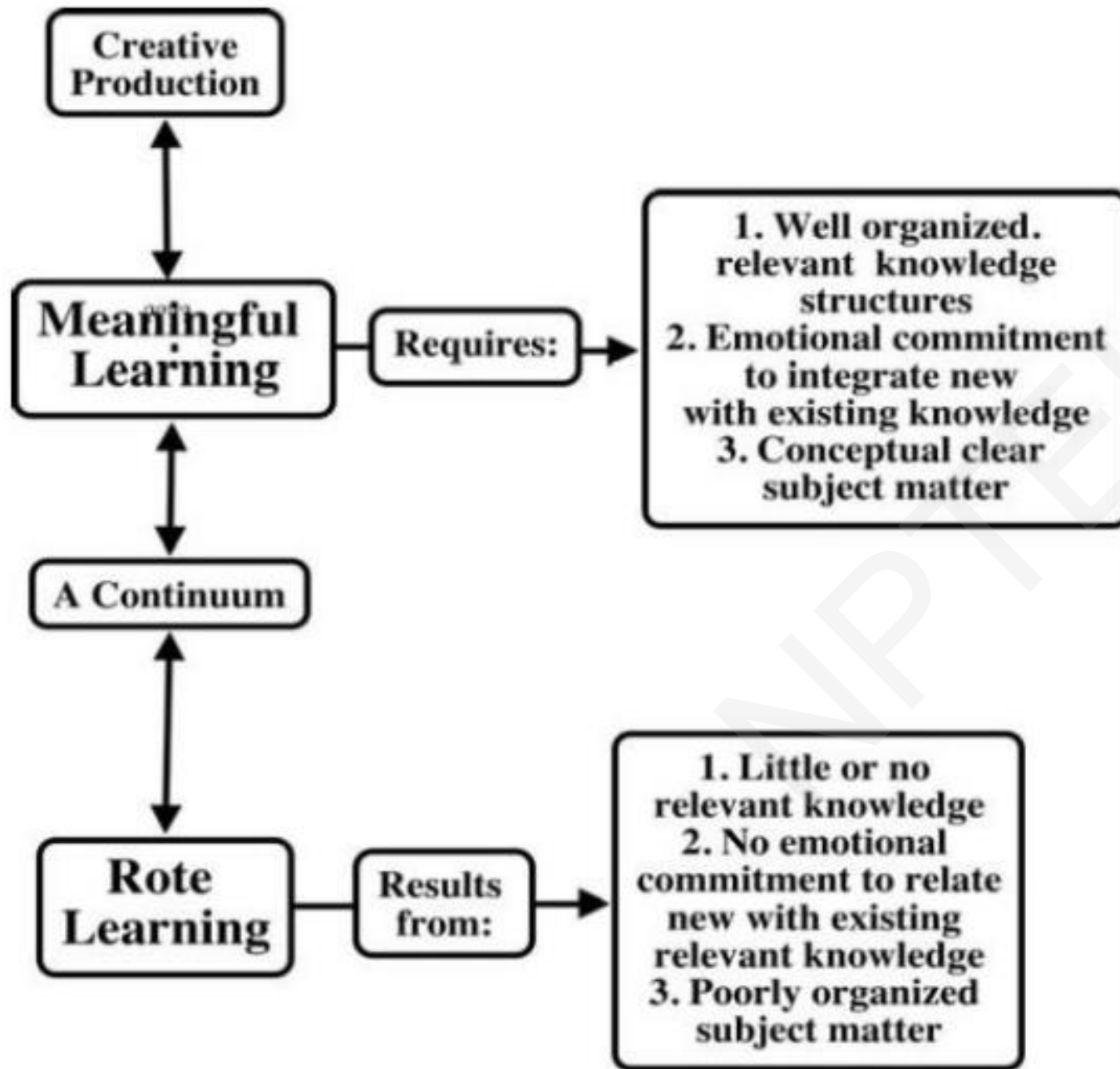
- Behavioral
- The behavioral model emphasizes **changing the visible behavior of the learner to be consistent with his or her own self concept.**
- As a result of its basis in the stimulus control/reinforcement theories, the behavioral model of instruction stresses that **learning tasks should be broken into a series of small, sequenced tasks** and behaviors.
- A unit of instruction might draw from several of the models, while a single lesson might incorporate aspects of more than one model.
- Instructional Strategies
- Decision making regarding instructional strategies requires **teachers to focus on curriculum**, the **prior experiences** and knowledge of students, **learner interests**, student **learning styles**, and the **developmental levels** of the learner.
- Such decision making relies on **ongoing student assessment** that is **linked to learning objectives** and processes.



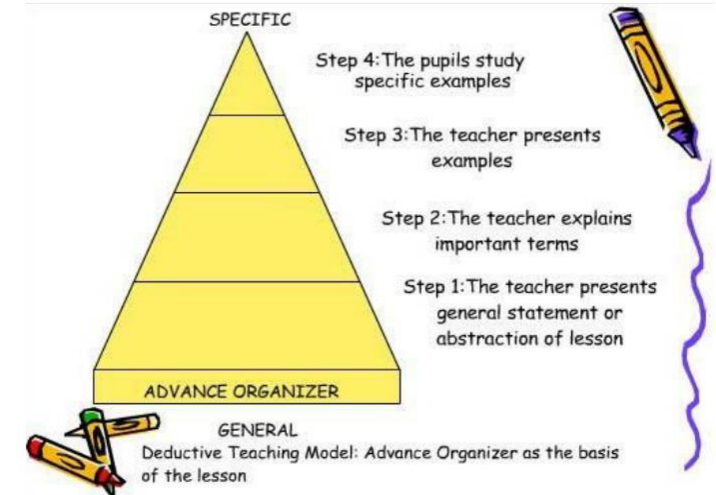


- **Expository Method of Teaching (David Ausubel)-**
- David Ausubel, an American psychologist, developed the learning theory known as **meaningful learning** in the 1960s.
- According to this theory, **meaningful learning** occurs when **new information is integrated into existing knowledge structures**, also known as **mental schemas**.
- Ausubel believed that **prior knowledge is the most critical** aspect of meaningful learning.
- He posited that **new information is learned best** when it is **linked to relevant existing knowledge**.
- He believed that **understanding concepts, principles**, and ideas are achieved through **deductive reasoning**.
- The most **important single factor** influencing learning is **what the learner already knows**.
- He developed the theory of meaningful learning and **advance organizers for effective teaching-learning process**.
- **Knowledge is hierarchically organized**; that new information is meaningful to the extent that it can be related to what is already known.





Deductive Teaching Model



- **Advance Organizers**

- Ausubel advocates the use of **advance organizers** as a mechanism to help **to link new learning material with existing related ideas**.
- Advance organizers are helpful in the way that they **help the process of learning** when difficult and **complex material** are introduced.
- This is satisfied through **two conditions**:
 - 1. The **student must process** and understand the **information presented in the organizer**-- this increases the **effectiveness** of the organizer itself.
 - 2. The organizer must indicate the **relations among the basic concepts** and terms that will be used.
- Advance organizers fall into **two categories**: **comparative** and **expository**.

- **Comparative Organizers**

- The main goal of **comparative organizers** is to activate existing schemas and is used **as reminders to bring into the working memory** of what you may not realize is relevant.
- A comparative Organizer is also **used both to integrate as well as discriminate**.



David Ausubel wrote :

- First, organizers provide advance ideational scaffolding.
- Second, they provide the learner with generalized overview of all major similarities and differences.
- Finally, they create an advance set in the learner to perceive similarities and differences.



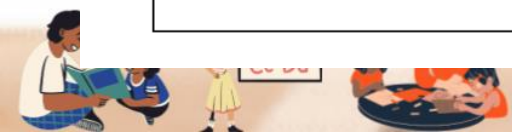
- **Expository organizers**
- Provide new knowledge that students will need to understand the upcoming information.
- Expository organizers are often used when the new learning material is unfamiliar to the learner.
- They often relate what the learner already knows with the new and unfamiliar material—this in turn is aimed to make the unfamiliar material more plausible to the learner.

Ausubel's Model of Meaningful Learning

<i>Phase One</i> <i>Advance Organizer</i>	<i>Phase Two</i> <i>Presentation of Learning task or Material</i>	<i>Phase Three</i> <i>Strengthening Cognitive Organization</i>
Clarify aim of the lesson	Make the organization of the new material explicit	Relate new information to advance organizer
Present the lesson	Make logical order of learning material explicit	Promote active reception learning.
Relate organizer to students' prior knowledge	Present material in terms of basic similarities and differences by using examples, and engage students in meaningful learning activities	

Advance Organizers

1. Lecture - effective form of teaching if:
 - a. ideas and concepts are organized.
 - b. ideas and concepts are connect to what learner already knows.
2. Advanced organizer
3. AO – three main purposes
 - Highlight key points.
 - Activate relevant knowledge.
 - Show relationships.





• Ausubel's Ideas-

- According to the **expository method** of teaching, the **learner is an active agent** who engages with and **interprets information** and **incorporates it into existing cognitive schemata**.
- The role of the **teacher** is not just to **present new information** but to do so **in a meaningful way** – taking account of the learner's prior experience.
- **New knowledge should always be subsumed under** (related to, integrated with) **previously familiar concepts**, a hierarchical way of organizing knowledge in mind, (**general ideas followed by more complex ones, general ideas form advance organizer, which is a general/subsuming framework for understanding new concepts**).
- A primary process in learning is **subsumption** (**in which new material is related to relevant ideas in the existing cognitive structure on a substantive, non-verbatim basis**).
- **Cognitive structures** represent the **residue of all learning experiences**; (**learning is based upon the kinds of superordinate, representational, and combinatorial processes that occur during the reception of information**).
- Thus, it's known **both as expository & subsuming teaching method**.

DAVID AUSUBEL

- DEVELOPED ADVANCED ORGANIZERS
- RELATE TO NEW CONTENT
- SUBSUMPTION THEORY
- PROGRESSIVE DIFFERENTIATION
- INTEGRATIVE RECONCILIATION
- GENERAL, ABSTRACT CONCEPTS TO NEW CONCEPTS





LEARNING & Instruction

Lecture : 35

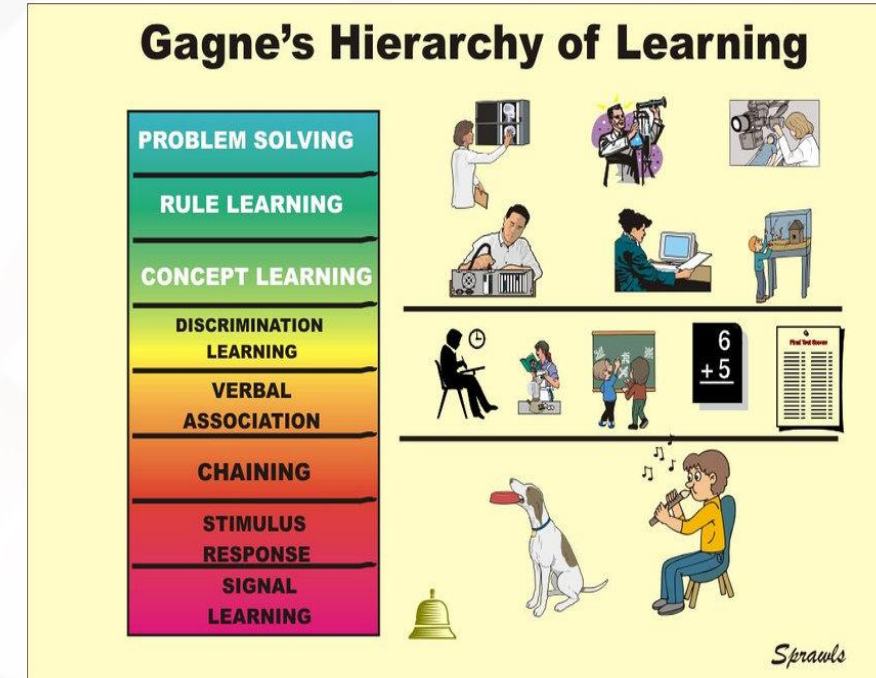


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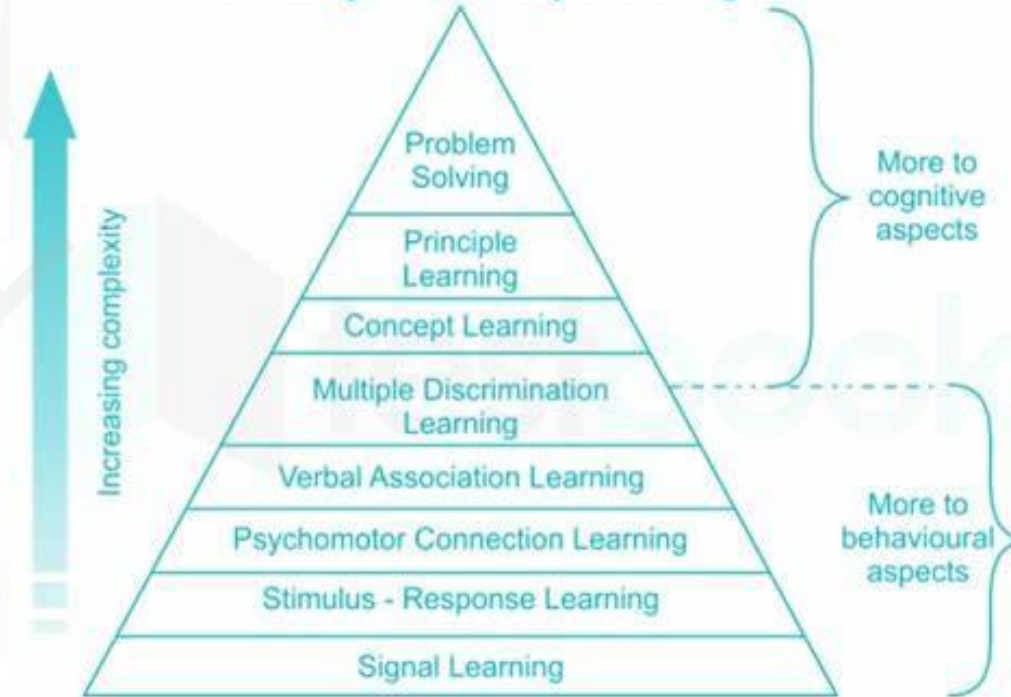
- **Robert Gagné's Theory of Learning & Instruction-**
- Robert Gagné is one of the behaviorist theorists; he took the best of both worlds, behaviorism and cognitivism, to create his view of instruction and its design.
- He believes that **learning results in behavior changes that are Observable**; called these changes in behavior outcomes.
- **Outcomes are descriptions of educational goals** in terms of what is to be accomplished through the prescribed learning activities.
- Gagné also maintains a **cognitive perspective** in which **learning was recognized as a result of an individual's cognitive efforts** to construct his or her personal knowledge.
- He focuses on memory and **how to plan curriculum in chunks so as not to overload memory**.
- His **instructional design models** reflect this **emphasis on thinking and remembering**.
- His theory divided into three elements-
- **Taxonomy of learning outcomes**
- **Conditions necessary to achieve the learning outcomes &**
- **Nine events of instruction designed to guide the teacher through the process of designing for learning**





- He identified **five categories of learning**; each require a different type of instruction in order for learning to occur verbal information, intellectual skills, cognitive strategies , motor skills and attitudes (**Taxonomy of Human learning capabilities**)
- His theory directed instruction that **combine the behavioral and information processing learning theories**;
- He believes that the **results of learning are measurable** through testing and that **drill, practice, immediate feedback or effective**.
- Each **learning outcome must have- pre-requisite knowledge** or skills must be identified;
- Instructor uses the **conditions of learning for the particular learning outcomes** to determine the conditions necessary for learning .
- Different **internal and external conditions** are necessary **for each type of learning**.
- The **teacher** must accomplish **at least three tasks-**
 - 1. Assure pre-requisite skills
 - 2. Supply instructional conditions
 - 3. Determine the type of learning

Robert Gagne's Hieracchy of Learning

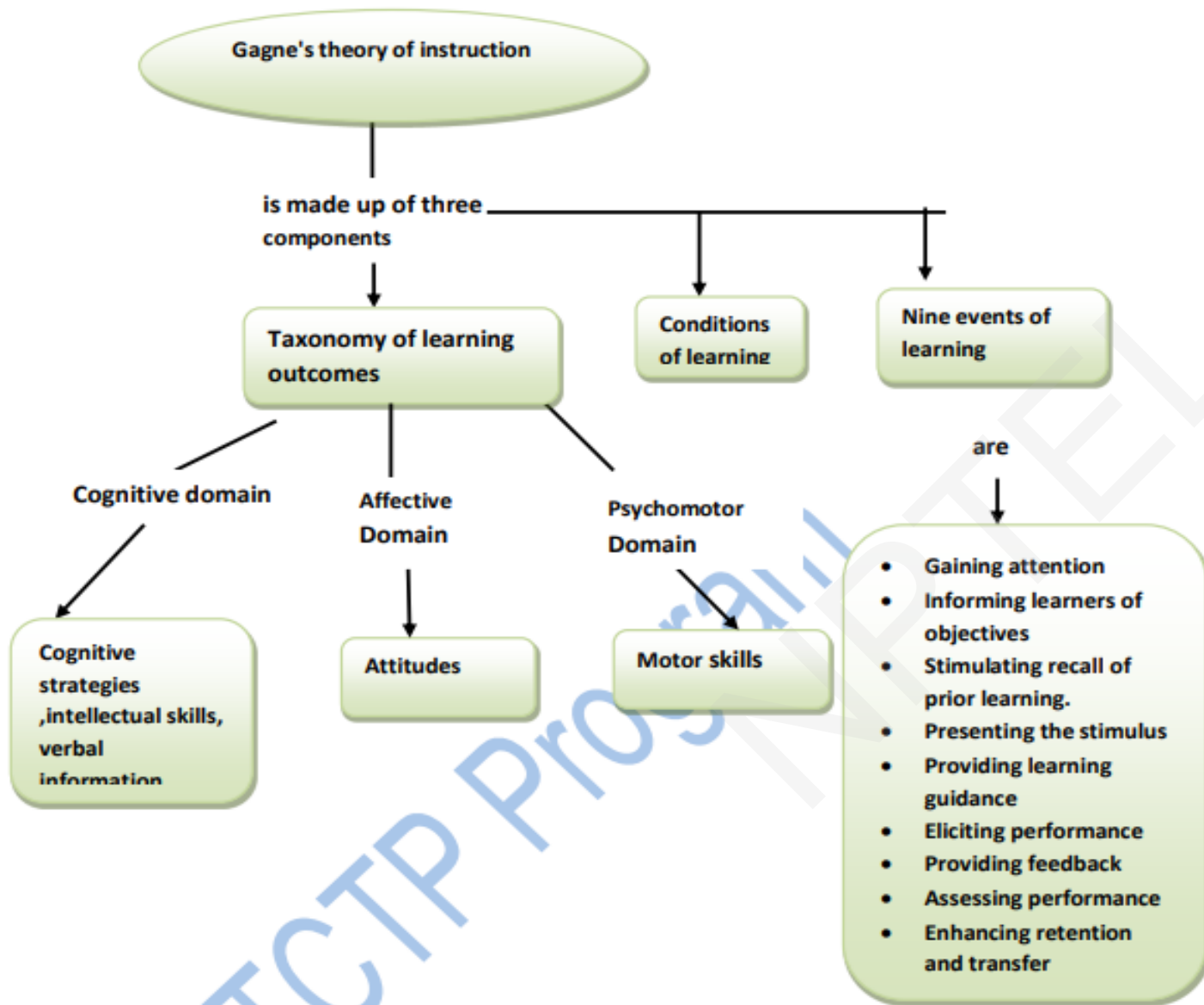


Types of Human Capabilities	Conditions	Principles for Instructional Events
Verbal Information	Retrieving stored information: the internal conditions to support this learning include • Preexisting of organized knowledge • Strategies for processing the new information	• Provide meaningful context of information for encoding • Provide elaborations, imagery, or other encoding cues • Organize information so that it can be learned in chunks
Intellectual Skills	Mental operations that permits individuals to respond to conceptualizations of the environment: • Discrimination • Concrete and defined concepts • Rule using • Problem solving: combining subordinate rules in order to solve a problem The internal conditions to facilitate this type of learning include: • Recalling prerequisite skills • Interacting in a variety of ways with the new learning • Applying the new skills to range and variety of different situations and contexts	• Provide varied concrete examples and rules • Provide opportunities for interacting with examples in different ways • Assess learners in new situations

Cognitive Strategies	An internal process by which the learners plans, controls, and monitors his/her own ways of thinking and learning, including • Task specific • General • Executive	• If task-specific, describe the strategy; if task general, demonstrate the strategy. • Provide opportunities for strategy specific practice with support and feedback
Attitude	An internal state, i.e. predisposition that affects an individual choice of action	• Provide respected models who enact positive behavior and reinforce the model • When learner enacts the behavior, provide reinforcement
Motor Skills	Capability to perform a sequence of physical movements. It involves three stages: • Learning the sequence of the movement • Practicing the movement • Refining the movement from the feedback from the environment	• Establish executive subroutine and provide for mental rehearsal. • Arrange several repetitions of skills with correct feedback



• GAGNE's LEARNING MODEL-



9 Events of Instruction

- #1 • Gain attention (the hook)
- #2 • Orient the learner (prep and objectives)
- #3 • Stimulate recall of prior knowledge
- #4 • Present content material
- #5 • Provide learner guidance
- #6 • Elicit performance "practice"
- #7 • Provide informative feedback
- #8 • Assess if lesson objectives have been learned
- #9 • Enhance retention and transfer



• Instruction Theory of Jerome Bruner: The Guided Discovery Learning

- Bruner sees the learner as an active agent; emphasizing the importance of existing schemata in guiding learning.
- Bruner argues that students should discern for themselves the structure of subject content – discovering the links and relationships between different facts, concepts and theories .
- Bruner (1966) hypothesized that the usual course of intellectual development moves through three stages: enactive, iconic, and symbolic, in that order.
- The aim of education should be to create autonomous learners (i.e., learning to learn).
- Cognitive growth involves an interaction between basic human capabilities and “culturally invented technologies that serve as amplifiers of these capabilities.”
- The culturally invented technologies include not just computers and television, but more abstract notions such as the way a culture categorizes phenomena, and language itself.

Discovery Learning

“Discovery learning is ‘an approach to instruction through which students interact with their environment – by exploring and manipulating objects, wrestling with questions and controversies, or performing experiments’.”



- Language serves to mediate between environmental stimuli and the individual's response.
- Modes of representation are how information or knowledge is stored and encoded in memory.
- Knowledge is represented and organized through different modes of thinking (or representation).
- He proposed three modes of representation:
 - Enactive representation (action-based)
 - Iconic representation (image-based)
 - Symbolic representation (language-based)
- He suggests it is effective when faced with new material to follow a progression from enactive to iconic to symbolic representation; this holds true even for adult learners.
- According to his theory a learner even of a very young age is capable of learning any material so long as the instruction is organized appropriately.

Bruner's Main Concepts

1. Representation

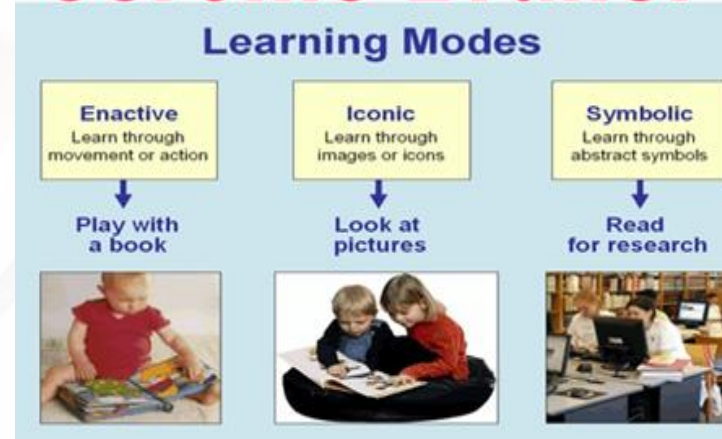
- a. Enactive representation – at the earliest ages, children learn about the world through actions on physical objects and the outcome of these objects.
- b. Iconic representation - learning can be obtained through using models and pictures.
- c. Symbolic representation – the learner has developed the ability to think in abstract terms.





- In the **Enactive mode**, (0 -1 year) –
- **Knowledge is stored** primarily in the form of **motor responses**.
- This mode is used within the **first year of life**; thinking is based **entirely on physical actions**, and **infants learn by doing**, rather than by internal representation (or thinking).
- It involves **encoding physical action-based information** and **storing it** in our memory.
- **Iconic Mode (1-6 years)**-
- **Information is stored as sensory images (icons)**, usually visual ones, like **pictures in the mind**.
- Also when we are learning a new subject, it is often helpful to have **diagrams** or **illustrations** to accompany the verbal information.
- **Thinking is also based on using other mental images (icons)**, such as hearing, smell or touch.
- **Symbolic Mode (7 years onwards)**-
- In this symbolic stage, **knowledge is stored primarily as language, mathematical symbols**, or in **other symbol systems**.

Jerome Bruner





JEROME SEYMOUR BRUNER
"LEARNING AS DISCOVERY"



- Symbols are flexible in that they can be manipulated, ordered, classified, etc. so the user isn't constrained by actions or images,
- According to Bruner's taxonomy, these differ from icons in that symbols are arbitrary.
- Language is important for the increased ability to deal with abstract concepts.
- He argues that language can code stimuli and free an individual from the constraints of dealing only with appearances, to provide a more complex yet flexible cognition.
- The use of words can aid the development of the concepts they represent .
- Education should aim to create autonomous learners (i.e., learning to learn).
- Bruner's concept of Discovery Learning – arguing that students should “not be presented with the subject matter in its final form, but rather are required to organize it themselves.....to discover for themselves relationships that exist among items of information”.





- The role of the **teacher should** not be to teach information by rote learning, but instead to **facilitate the learning process**.
- A good **teacher should** design lessons that **help students discover the relationship between bits of information**.
- This requires the **teacher to be actively involved in lessons**; providing **cognitive scaffolding** which will **facilitate learning** on the part of the student.
- **Scaffolding refers** to the steps taken **to reduce the degrees of freedom** in carrying out some task so that the **child can concentrate on the difficult skill** she is in the process of acquiring (Bruner, 1978).
- Bruner states that the **level of intellectual development determines the extent to which the child has been given appropriate instruction** together with practice or experience.
- He views **symbolic representation as crucial for cognitive development**, and since language is the primary means of symbolizing the world, he attaches great importance to **language in determining cognitive development**.

Bruner's Instructional Scaffolding

What is Scaffolding?

An instructional scaffold is a temporary learner support provided to help a student learn how to complete a task. The scaffold should be withdrawn once the student is capable of completing the task alone.

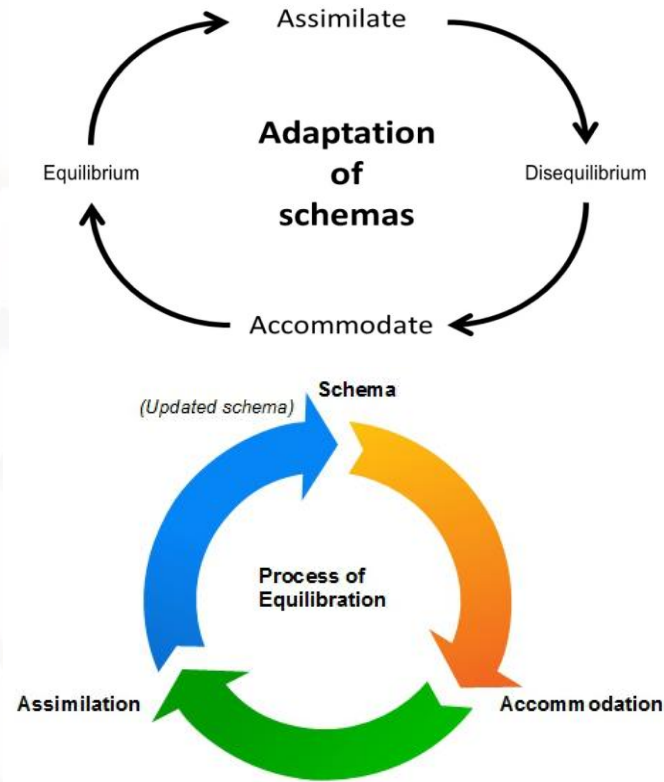
Examples of Scaffolding

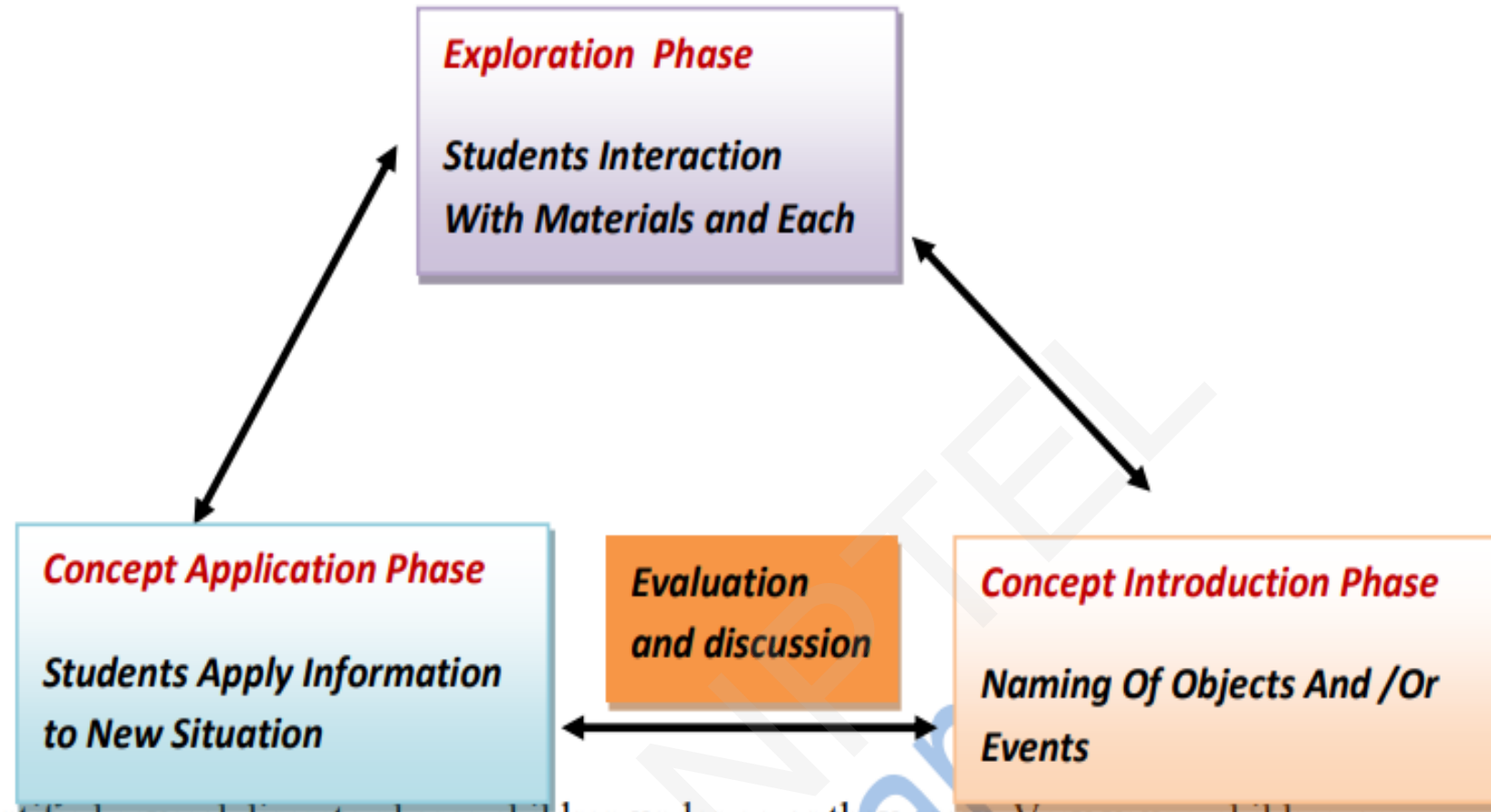
Modelling	Demonstrate to the student how to complete the task before asking them to give it a go themselves. This will give them a chance to observe then try out good practice.
Directing Attention	Intervene to direct students' attention to important elements of a task if you feel the student is going off track. However, remember that following bad leads can sometimes also be a good lesson.
Open-Ended Questioning	Open-ended questions require students to respond in full sentences, think deeply about their answer, and justify their response.
Breaking the Task into Smaller Parts	If a task is complex or overwhelming, give students a series of small steps toward success. Once the student has completed each small step, ask them to do the task from beginning to end without support.
Use Visual Aids	Visual aids with advice or prompts can act as cues to help students work through their task. Consider placing visual aids strategically around the classroom for students to draw upon when necessary.





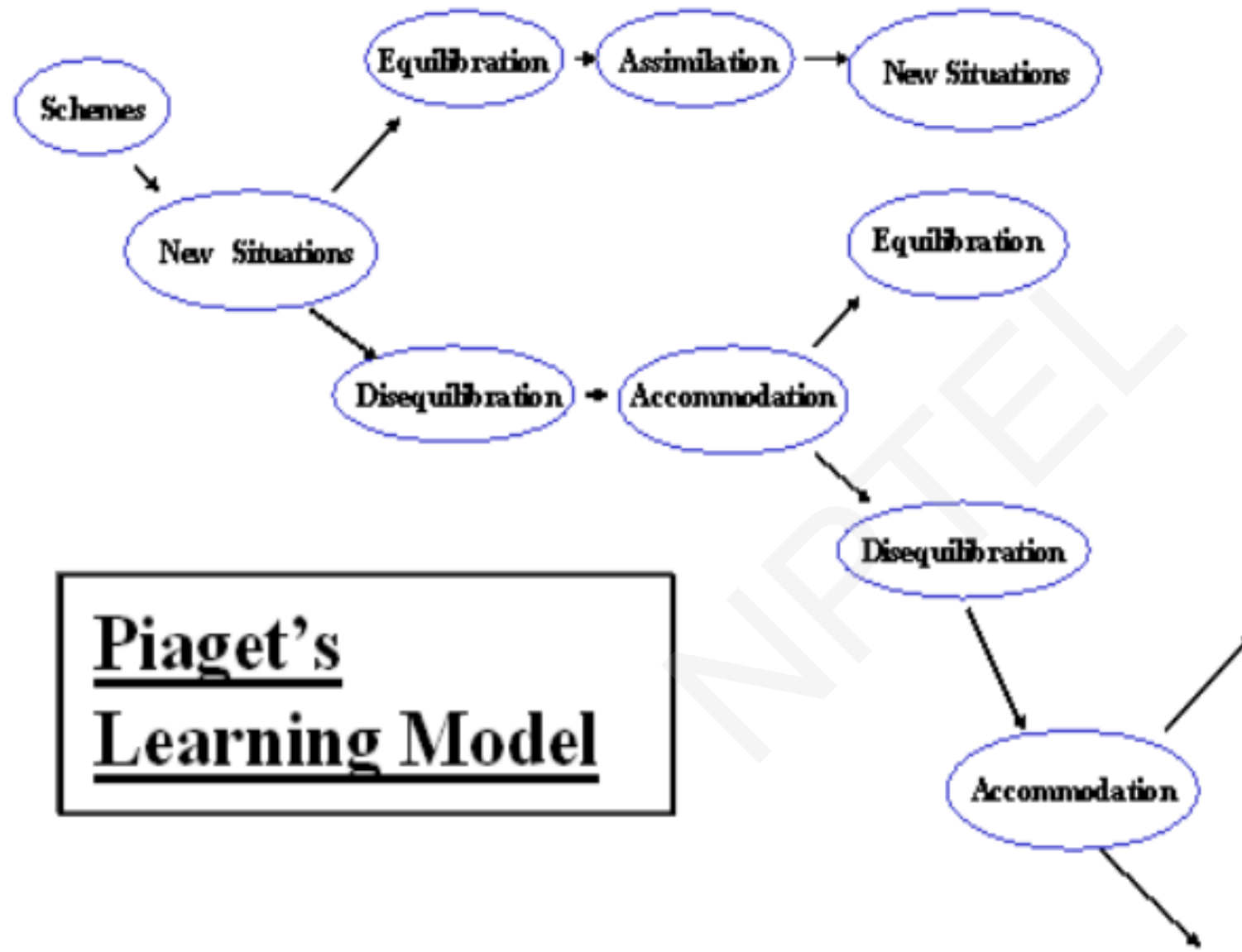
- For **Jean Piaget** learning is cyclical. Piaget's learning cycle includes an **exploration, concept introduction** and **concept application** phase;
- Humans learn by constructing their own knowledge.
- The primary role of **teacher should be to motivate the children to create their own knowledge** through their personal experiences.
- Teachers are merely **facilitators who support learners** in the learning process.
- **Learners should take control in making decisions** in line with their needs and cognitive state.
- Piaget **advocated non intervention** saying that everything one teaches a child prevents him from inventing or discovering.
- Children develop **knowledge through active participation** in their learning ;
- Learners will **build their own knowledge through experience**.
- This experience will **help them build mental models** which they will further **develop through assimilation and accommodation**.





He identified several discrete phases children under go as they grow. Very young children are only able to process concrete operational information they are unable to think abstractly, reflect on the past or project into the future. Older children develop these abilities over time.





Piaget's Learning Model



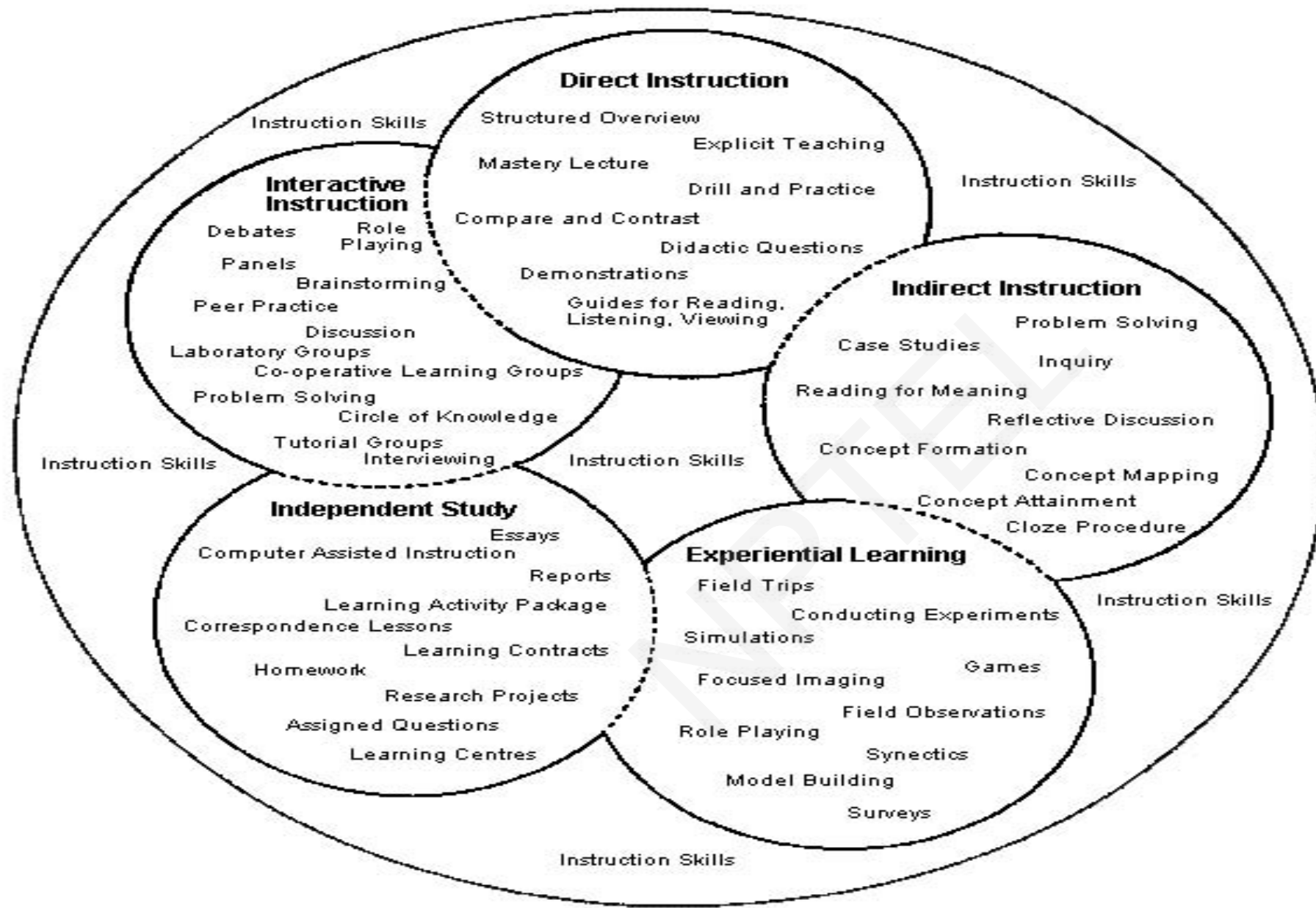


Figure 5. Instructional Strategies



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Thank
you

